Geospatial in USD

Proposal quality and working evidence

The exercised workflows revealed implementation defects and dataset limitations, rather than a new missing data-model decision. A working image alone would not establish that conclusion: the checks trace authored facts to required behavior, independent expected answers and visible failures.

The review candidate defines position and physical model attitude separately from ordinary transforms, keeps WKT authoritative, and specifies composition, interpolation, native-data association, Profiles and explicit baking. Those detailed conventions still require author approval. Implementation success does not establish group adoption or complete conformance.

Fresh source generation from the complete proposed contract. 15 October 2026.

Authored inputs and computed results

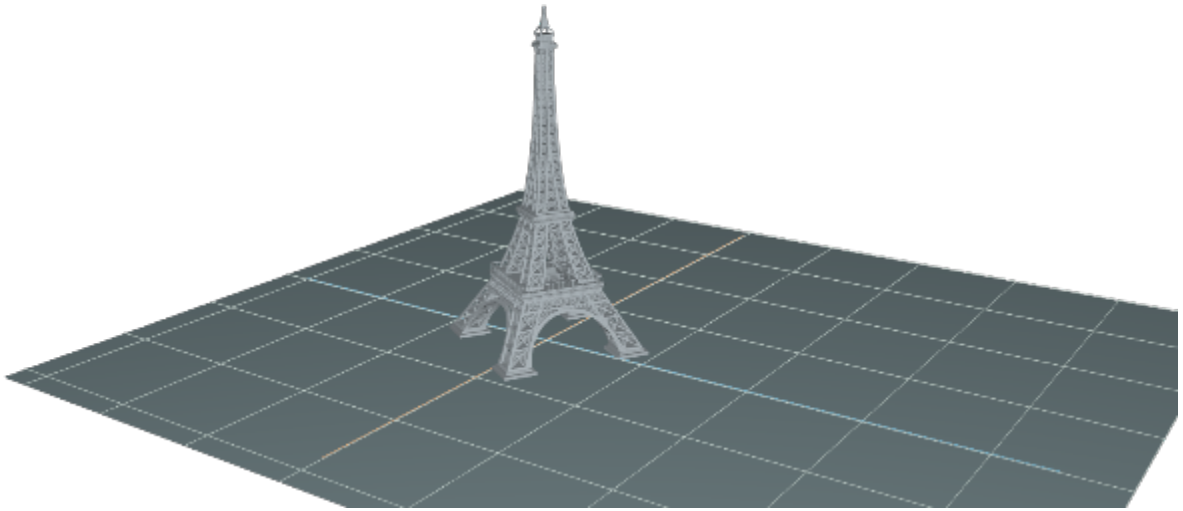
An author records what the source means and how the model is placed. Consumers compute projected positions, convergence, scale, orientation and bounds. Those computed results are runtime data, rather than additional properties on the source.

Authored input	Meaning
uniform token crs:wkt	One complete shared CRS definition, including its coordinate units and vertical reference.
double3 crs:position	Absolute 3D model origin in the source CRS's semantic component order.
double3 crs:orientation	Physical heading, pitch and roll in degrees against geodetic east/north/up.
Ordinary USD xformOps	Local asset transforms and intentional project adjustments, in their specified contexts.
External-data association	Native asset, format, selected field and coordinate domain. Values and times stay associated with their original samples.

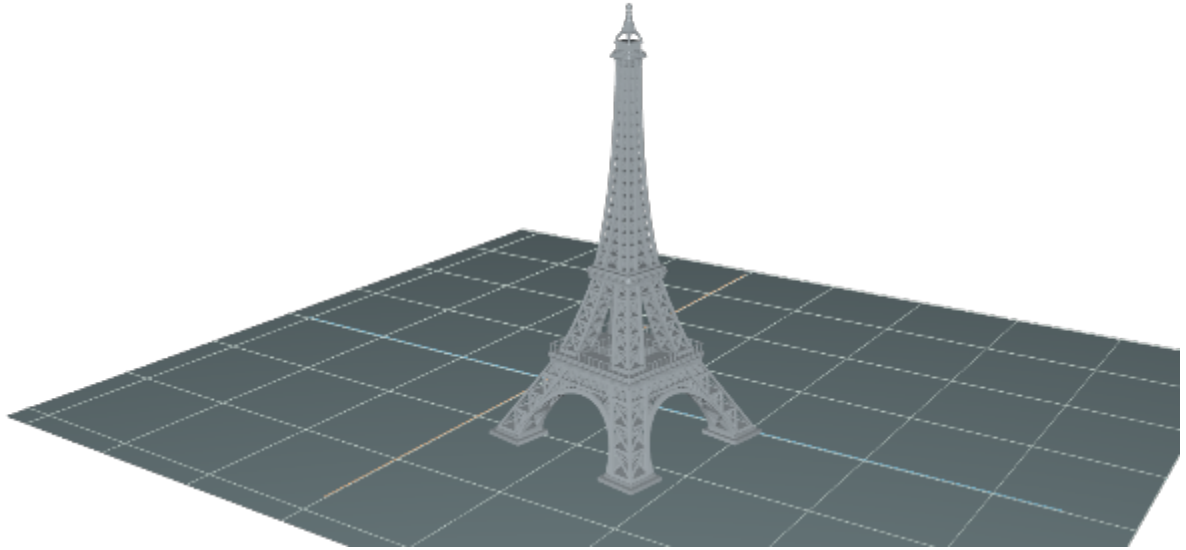
Computed projection effects stay out of the source schema.

Placement and ordinary USD adjustments

The tower first receives its physical attitude and CRS placement. The anchor's ordinary adjustments then act in the fixed working chart. Descendant transforms keep their model-local meaning, so a child offset follows the model's attitude.



30° true-north heading, nominal 300 m



Scale 1.15, grid rotation 20°, east 120 m, south 60 m

Same camera and 100 m display grid. Source position and physical attitude stay unchanged.

Independent survey controls

Both readers reproduce the supplied France and Colorado coordinate tables within 0.101 mm in these cases. The acceptance distance was 2 mm before execution. The provider tables supply the expected answers independently of either implementation.

Coordinate comparison	Rows	Largest Python error (mm)	Largest C++ error (mm)
France geographic to geocentric	59	0.100	0.100
France CC49 to Lambert-93	59	0.061	0.061
Colorado state plane to calibrated site grid	5	0.034	0.034

Provider coordinates are the expected answers. Both readers use PROJ, so their agreement is not an independent geodetic algorithm check.

USD Profiles and adoption

USD Profiles lets a consumer learn whether an asset requires geospatial resolution before traversing or loading its payloads. The publisher maintains the conservative declaration. A successful geometry bake can instead declare that ordinary USD is sufficient.

```
prepend apiSchemas = ["ClaimsAPI"]
customData = {
    dictionary profilesInfo = {
        dictionary capabilityUsages = {
            string "usd.geospatial.crsResolution.v1" = "hard"
        }
        dictionary profileCompatibility = {
            string[] "usd.geospatial.publication.v1" = []
        }
    }
}
```

Existing ClaimsAPI and customData carrier. Versioned identities remain proposed.

Profiles before payload loading

The publisher maintains conservative coverage. The consumer reads the interface without traversing or loading content.

Declaration	Consumer conclusion
Hard geospatial usage	Resolution is required, including while content is unloaded.
Qualified complete summary, empty usages	No retained content requires geospatial resolution.
No hard usage and no complete summary	Coverage is unknown. Absence alone proves nothing.

A stronger omission cannot conceal retained dependency. The publisher must refresh or withdraw a stale completeness claim.

Technical registration and schema implication are tested. Namespace stewardship and approval ownership remain review items.

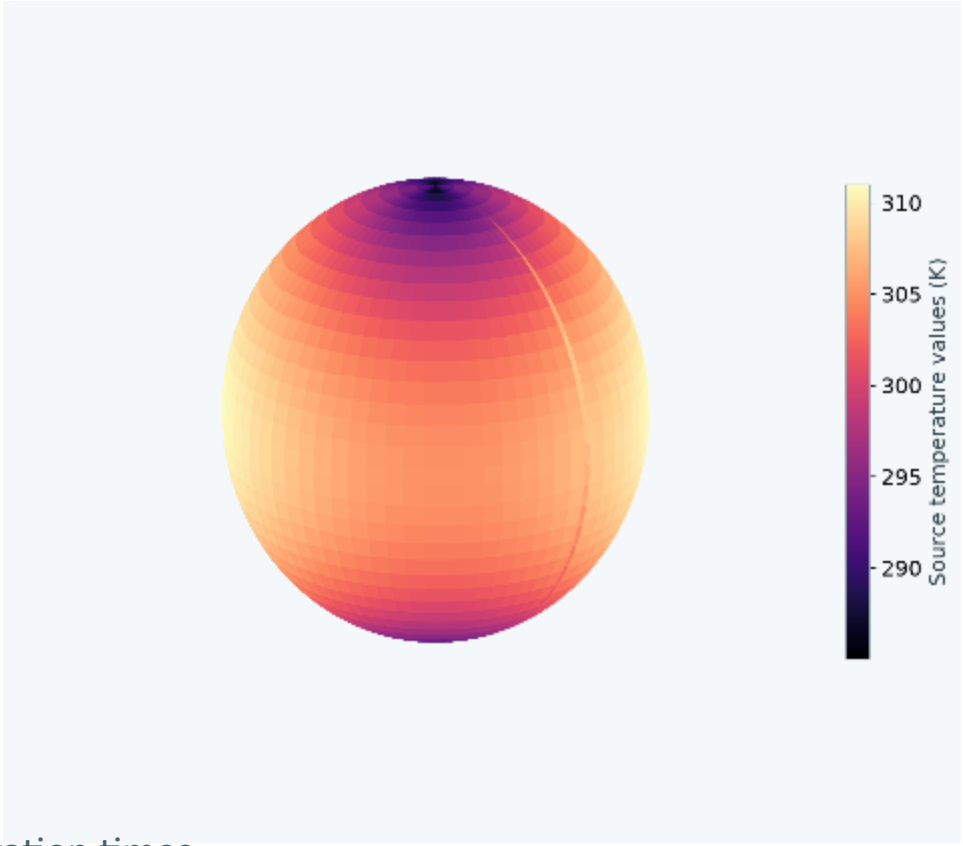
Measurements without a renderer

The same coordinate interpretation supports analysis as well as visualization, from project-sized data to global domains. Measurements retain their values, sample identities, masks and observation times while their locations resolve into the requested CRS.

2,664 climate demonstration values

5,328 illustrative samples across two declared times

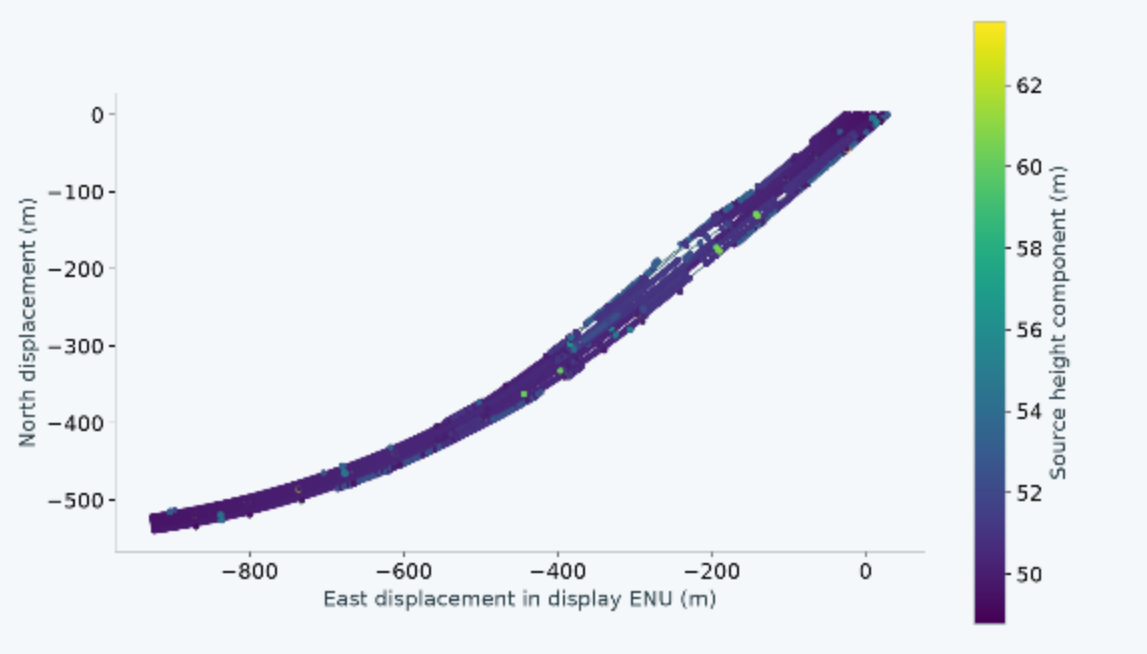
Declared CF metadata, units and synthetic height. Observational provenance is unverified.



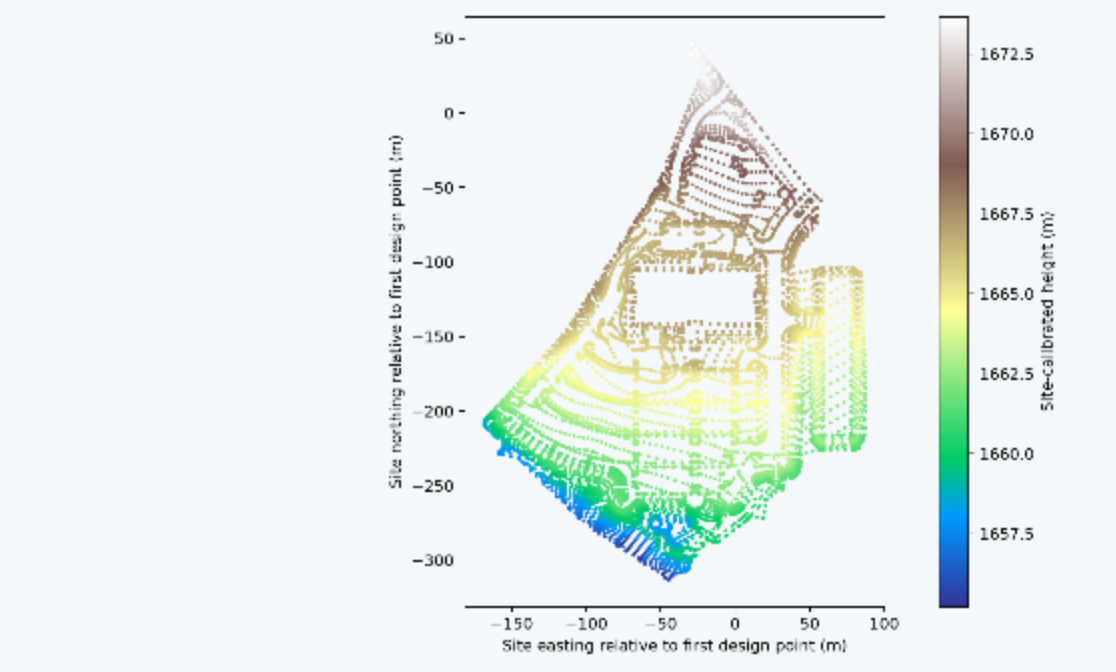
Both readers preserve sample identities, values, masks and observation times.

Railway and calibrated construction data

The original railway and Westminster design data retain their coordinate meaning across the two readers. The run checks 15,822 railway positions and 13,899 construction design-breakline positions, alongside the independent survey controls.



15,822 railway positions
GeoJSON geometry and height semantics



13,899 design-breakline positions
Explicit derived CF copy of LandXML

Original files remain unchanged. The complete site-calibration WKT retains horizontal and vertical conversions.

Two readers and consuming applications

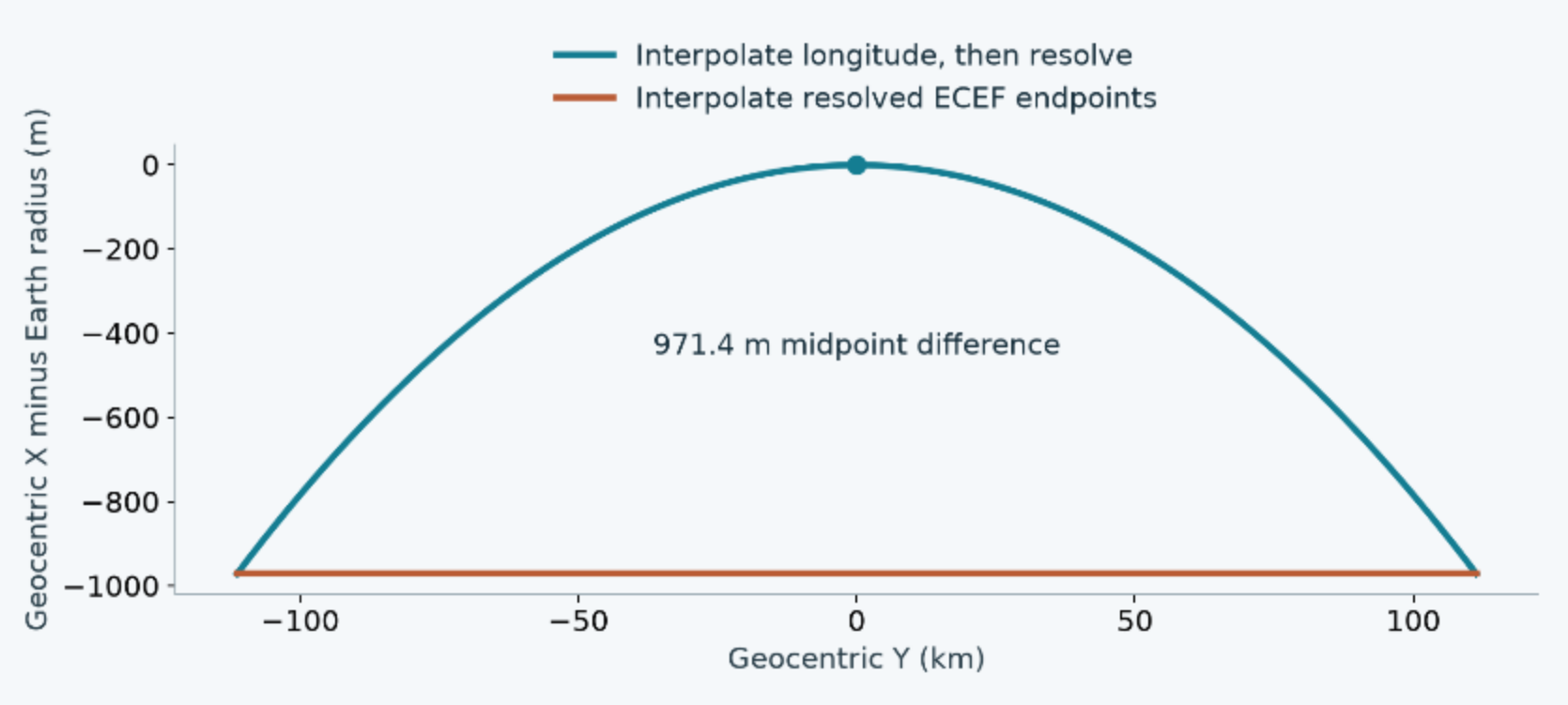
Python and native C++ independently read composed USD placement inputs. Omniverse consumes the Python-derived geometry through an actual application context stage. Hydra Storm consumes the C++ ordinary bake. These are two placement implementations and two consumer paths.

Executed path	Evidence
Python reader	Resolves the original composed source and produces coordinates and an ordinary bake.
Native C++ reader	Reads USD and calls PROJ directly; it does not invoke Python placement.
Omniverse stage write/readback	All 163,416 Eiffel vertices match the supplied Python-derived ordinary geometry, with zero write/read difference.
Hydra Storm	Successfully renders the C++-derived ordinary geometry shown above.

Eiffel bake difference: at most 0.00763 mm. Omniverse stage write/readback: 0 m.

Source interpolation before projection

The equatorial control places the source-interpolated midpoint on the surface. Interpolating only projected endpoints would miss by 971.4 m.



A bake records actual sample keys and time-code scale. Those keys alone do not certify the original between-sample trajectory.

Meaningful failures and coverage limits

The prototype rejects missing height, invalid bindings, blocked or wrongly typed inputs, unsupported axes, missing resources and epoch-dependent transformations. It also rejects an ambiguous 180° interpolated pose while retaining exact-sample and held evaluation.

Requested behavior	Prototype disposition
Continuous nonlinear face, curve or subdivision bounds	Unsupported without a domain certificate. Vertex tests provide finite-point evidence.
Geometric point/vector/normal primvar export	Explicitly rejected. Ordinary mesh normals have derivative-based transport.
Native format coverage	GeoTIFF IFD 0 and the declared CF unit/role subset. Other combinations need separate verification.
General inverse and composition coverage	The tested controls pass. Global inverse uniqueness and all value-clip combinations remain unverified.

These are implementation or verification limits against an explicit contract. Complete conformance is not established.

Review conclusion

The tested proposal contracts can drive a fresh implementation without inventing authored facts. Remaining review should focus on approving the concrete model/runtime conventions and Profiles ownership, while keeping prototype limitations separate from proposal completeness.

Coordinate epochs remain outside the initial scope, with a path for future representation and motion-model support. This run supplies no default height or epoch, no crs:scale, and no computed geospatial properties on source content. Larger-scale geographic data remains part of the demonstrated scope.

The assessment supports review of the concrete data model and runtime conventions. Profiles ownership and prototype coverage remain explicit.

Eiffel: SDC PERFORMANCE, CC BY 4.0. Illustrative placement, scale and display-material changes.